

OPERATORS IN C

Agenda

- Concept of Operators
- Classification of operators
- References

Classification

- ✓ Arithmetic operators
- ✓ Conditional operators
- ✓ Bitwise operators
- ✓ Relational operators
- ✓ Logical operators
- ✓ Assignment operators
- ✓ Increment and decrement operators
- ✓ Special operators

Arithmetic Operators

- ✓ Integer Arithmetic
- ✓ Real Arithmetic

Integer Arithmetic

Operator	Name
+	Addition
-	Subtraction
*	Multiplication
/	Division
%	Remainder

If $a = 15$ and $b = 2$

$$a - b = 13$$

$$a + b = 17$$

$$a / b = 7$$

$$a \% b = 1$$

$$a * b = 30$$

Real Arithmetic

- Involves decimals.

If x,y are floats

$$x=6.0/7.0=0.857143$$

$$y=-2.0/3.0=-0.666667$$

Conditional operators

- The conditional operator consists of 2 symbols the question mark (?) and the colon (:)
- Syntax:
exp1 ? exp2: exp3

Example

```
a=16  
b=25;  
x=(a>b) ? a : b;
```

```
if (a>b)  
    x=a;  
else  
    x=b;
```


Bitwise operators

Operator	Meaning
&	Bitwise AND
	Bitwise OR
^	Bitwise Exclusive OR
<<	Shift left
>>	Shift right

Result of logical Bitwise Operation

op1	op2	op1&op2	op1 op2	op1^op2
1	1	1	1	0
1	0	0	1	1
0	1	0	1	1
0	0	0	0	0

Example

Bitwise AND

x - - -> 0000 0000 0000 1101

y - - -> 0000 0000 0001 1001

x&y- - -> 0000 0000 0000 1001

Bitwise OR

x - - - > 0000 0000 0000 1101

y - - -> 0000 0000 0000 1001

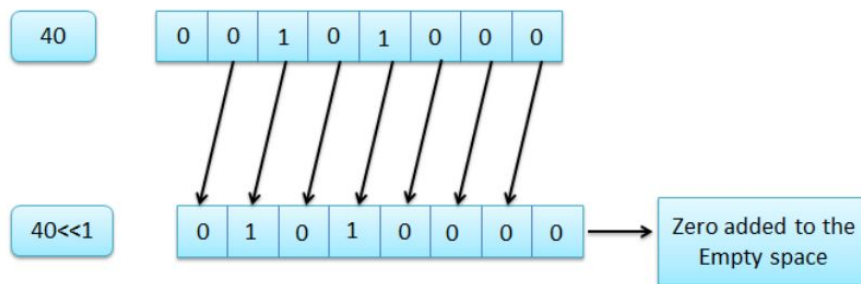
x|y - - -> 0000 0000 0000 1101

Bitwise Exclusive OR

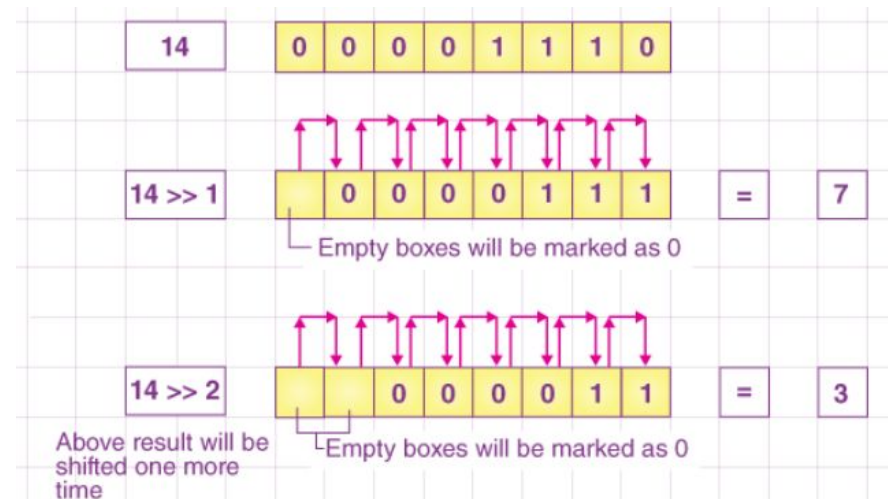
x - - ->	0000	0000	0000	1101
y - - ->	0000	0000	0001	1001
x^y - - ->	0000	0000	0001	0100

Bitwise shift operators

Left shift $op \ll n$



Right shift $op \gg n$



Bitwise Complement Operators

~ one's complement operator is a unary operator.

Input: ~ 0000 0011

Output: 1111 1100

Relational Operators

Operator	Meaning
<	is less than
<=	is less than or equal to
>	is greater than
>=	is greater than or equal to
==	is equal to
!=	is not equal to

Logical operators

Operator	Meaning
&&	Logical AND. True only if all operands are true
	Logical OR. True only if either one operand is true
!	Logical NOT. True only if the operand is 0

Assignment Operators

Used to assign a value to a variable

Example

$x = a + b$

Increment and Decrement

1. ++variable name
2. variable name++
3. --variable name
4. variable name--

x= 5;

y = ++x; (prefix)

In this case the value of y and x would be 6

x= 5;

y = x++; (post fix)

Then the value of y will be 5 and that of x will be 6.

Reference

- <https://www.programiz.com/c-programming/c-operators>

THANK YOU